
Calibrating What Behavior Can Identify: Conformal Reward Sets for Inverse Reinforcement Learning

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Abstract

1 Inverse reinforcement learning (IRL) returns a reward, and a policy is then op-
2 timized against it as if it were correct. Conformal inverse optimization (CIO)
3 offers a principled alternative for one-shot inverse problems: calibrate a radius
4 around the point estimate on held-out decisions, then optimize robustly over the
5 resulting set. We ask what survives the move to IRL. The conformal step transfers
6 untouched; two others break, both because behavior pins down a reward only up
7 to transformations that leave every policy value unchanged. First, CIO scores a
8 demonstration by an angle between reward vectors, which depends on coordinates
9 the problem never fixes: a feature reparameterization that changes no reward and
10 no demonstrator’s policy still moves the calibrated angular radius by 24° on av-
11 erage. Scoring instead by the spread a reward difference induces over achievable
12 policy values restores invariance and keeps the score a linear program. Second,
13 which event a quantile certifies is decided by which distance to the cone of rewards
14 explaining new behavior is scored: the nearest point gives intersection with that
15 cone, the farthest point of the normalized cone gives containment of the latent
16 reward. Only the first is a linear program, and at a 0.80 target we measure the two
17 events at 0.82 and 0.44. We give the enlarged radius that prices the relaxation, and
18 the point past which the resulting robust program provably says nothing.

19 1 Introduction

20 IRL turns observed behavior into a reward estimate, and that estimate is then optimized to pick
21 a policy. Uncertainty is dropped exactly where it matters, because two rewards that explain the
22 demonstrations equally well can recommend different actions. Conformal inverse optimization [Lin
23 et al., 2024] is an appealing remedy for the analogous one-shot problem: observe a population of
24 decision makers, fit a point estimate on part of it, and on the rest record how far that estimate sits
25 from the objectives that would rationalize each held-out decision. The conformal quantile of those
26 distances defines an uncertainty set with a finite-sample guarantee, and a robust optimization over
27 that set replaces the point prescription. The recipe is modular and looks portable. This paper reports
28 what happens when it is carried into IRL, component by component (Table 1).

29 One component transfers for free. Once per-demonstrator scores exist, the calibration arithmetic—
30 rank them, take the $\lceil \gamma(N + 1) \rceil$ -th, invoke exchangeability—is split conformal prediction exactly
31 as in Vovk et al. [2005] and Lin et al. [2024]; we inherit it wholesale. Sequential decision making
32 changes nothing about that argument. The work is in the two components on either side of it: which
33 distance gets calibrated, and what the resulting event is good for.

Table 1: The CIO recipe in IRL. Bold rows mark what we had to change; the rest, including the conformal quantile itself, are inherited.

CIO ingredient	Counterpart here	Status
Point estimate from decision data	any IRL fit (MaxEnt, Bayesian, ...)	inherited
Inverse-feasible set of a decision	Bellman cone $K(\pi)$, linear in θ	inherited
Score: angle to the estimate	span distance to $K(\pi)$; invariant, still an LP	redesigned (§4.2)
Split-conformal quantile	unchanged	inherited
Calibrated event $\Rightarrow$ parameter set	convex relaxation of a nonconvex sup-score, priced by a cone-width bound η	relaxed + priced (§4.3)
Robust decision	reference-relative robust MDP, one LP	redesigned (§4.4)

The score is measured in the wrong space. CIO scores a demonstration by the angle between the point estimate and the nearest objective that rationalizes it. In IRL that angle lives in coordinates the problem does not fix. A reward vector can change without changing the difference between any two policies’ values: add a potential-shaping term, or alter the reward on a state no policy reaches. The angle registers such a change; behavior cannot. An uncertainty set built from angles can therefore hit its target rate while much of its radius covers differences no demonstration could exhibit. This is not hypothetical: we apply an invertible feature reparameterization that leaves every scalar reward and every demonstrator’s policy pointwise unchanged, and the calibrated angular radius still moves by 24° on average (§5). Two identical problems get two different uncertainty sets. The fix is not a better estimator but a prior question: what does behavior actually distinguish? Everything observable here factors through the achievable occupancy measures, so we measure a reward difference v by the spread of policy values it produces, $\|v\|_{\mathcal{D}} = \max_{\mu} V_v(\mu) - \min_{\mu} V_v(\mu)$, a seminorm whose null space is exactly the directions behavior cannot see.

Intersection is not containment. The second break matters more. Split conformal certifies that *some* reward in the calibrated set makes the new demonstrator optimal; a robust prescription needs the demonstrator’s *own* reward to lie in the set. In one-shot inverse optimization the two are close; in IRL, where one policy is rationalized by an entire cone, they are not. At target 0.8 we measure the first event at 0.82 and the second at 0.44, and no amount of calibration closes that gap. The two events differ by which distance to the cone is calibrated. Scoring the distance to the *nearest* point—the quantity a linear program can compute—certifies intersection. Scoring the distance to the *farthest* point of the normalized cone certifies containment, under no further assumption, but is a nonconvex maximization. Our construction is the convex relaxation together with an explicit account of what it costs: a factor of two from renormalizing the witness, plus a certified bound η on the cone’s width, giving $R = \min\{2, 2q_\gamma + \eta\}$. With no side information, $\eta = 2$, and the resulting guarantee is then exactly vacuous rather than merely loose.

What we report is diagnostic rather than a performance improvement: half of CIO’s pipeline moves to IRL untouched, two components must be rebuilt in decision space, and the step from intersection to containment is a relaxation whose price we state rather than hide. Regret against point-estimate IRL is therefore not the yardstick, and §5 reports where our construction does and does not lower it.

2 Related work

This work sits between two literatures. Inverse optimization recovers objectives from observed decisions [Ahuja and Orlin, 2001, Chan et al., 2019], and CIO adds finite-sample calibration to it [Lin et al., 2024]; this is a different problem from data-driven robust optimization, where the coefficients are observed or predicted rather than latent [Chenreddy et al., 2022, Sun et al., 2023]. IRL has known since Ng and Russell [2000] that behavior determines a reward only up to a cone, and recent work characterizes those feasible sets and asks what extra structure shrinks them—experts of known suboptimality under a generative model [Poiani et al., 2024], a unified LP view of demonstrations and feedback [Kim et al., 2024], or an account of what behavior provably cannot resolve [Skalse et al., 2023] and what that costs a downstream decision [Lazzati and Metelli, 2025, Kim et al., 2026]. That

line attaches no finite-sample probability to its set, and CIO never asks whether its geometry still means anything when the parameter is identified only up to a cone. We ask what happens in between. For the geometry we borrow rather than reinvent. EPIC canonicalizes shaping away [Gleave et al., 2021], DARD restricts to realizable transitions [Wulfe et al., 2022], and STARC unifies both, showing that any metric with matching worst-case-regret upper *and* lower bounds is bilipschitz equivalent to a STARC metric [Skalse et al., 2024]. Our occupancy span plays the role of a canonicalization–normalization pair in the same spirit, taken relative to the known P , so we propose no new reward metric; what is new is using this geometry as a conformal score over Bellman cones, and being explicit about the identification assumption the resulting radius still needs. On the robust side, tractability is classically recovered by rectangularity [Iyengar, 2005, Nilim and El Ghaoui, 2005, Wiesemann et al., 2013], and Gadot et al. [2024] show that non-rectangular reward ambiguity is tractable in occupancy space—the space we also work in, with the difference that their ambiguity set is given rather than calibrated. Conformal prediction has appeared in RL for other purposes [Gupta et al., 2023, Pan et al., 2026, Zhou and Zhu, 2025], but we are not aware of prior work that calibrates an inverse-feasible reward set from a population of IRL demonstrators.

3 Setup

A tabular MDP $M = (S, A, P, \rho_0, \beta, \phi)$ has known kernel P , initial distribution ρ_0 , discount $\beta \in [0, 1)$, and features $\phi : S \times A \rightarrow \mathbb{R}^d$ inducing $r_\theta(s, a) = \phi(s, a)^\top \theta$. A population of demonstrators draws rewards $\hat{\theta}_1, \dots, \hat{\theta}_N \sim \Lambda$ i.i.d.; demonstrator k acts optimally under $\hat{\theta}_k$, producing a deterministic policy π_k that we assume is observed exactly rather than estimated from finite trajectories. Let $M(P)$ be the polytope of normalized discounted occupancy measures, Φ the $(|S||A|) \times d$ feature matrix, and $V_\theta(\mu) := \theta^\top \Phi^\top \mu / (1 - \beta)$. The rewards that make an observed policy optimal form the cone

$$K(\pi) := \{\theta : Q_\theta^\pi(s, a) \leq Q_\theta^\pi(s, \pi(s)) \forall s, a\}.$$

Substituting the Bellman resolvent removes the unknown value function and makes $K(\pi)$ exactly $|S|(|A| - 1)$ halfspaces in θ alone (Proposition 5), which is the sequential restatement of Ng and Russell’s feasible-reward characterization.

Assumption 1 (Exchangeable calibration unit). *The pairs $(\hat{\theta}_k, \pi_k)$, $k = 1, \dots, N_{\text{val}}, N_{\text{val}} + 1$, are i.i.d. (hence exchangeable), and M —in particular P —is shared and fixed across all demonstrators and the deployment instance.*

Each demonstrator therefore plays the role of one (decision, context) pair in CIO’s setting; per-trajectory exchangeability within a single demonstrator would instead be broken by within-trajectory Markov dependence. The pipeline is Table 1 read top to bottom: split the population, fit $\hat{\theta}$ on the training split with any IRL method, calibrate on the rest—write N_{val} for the size of that split—convert intersection into containment, and solve a robust MDP. The inherited CIO baseline uses the spherical cap $C(\bar{\theta}, \alpha) = \{\theta : \|\theta\|_2 \leq 1, \theta^\top \bar{\theta} \geq \cos \alpha\}$ (Appendix A); that specific choice, not the calibrate-then-robustify idea, is what reward non-identifiability damages.

4 Calibrating in decision space

4.1 A distance that behavior can see

For $v \in \mathbb{R}^d$, define

$$\|v\|_{\mathcal{D}} := \max_{\mu \in M(P)} V_v(\mu) - \min_{\mu \in M(P)} V_v(\mu) = V_v^*(\rho_0) + V_{-v}^*(\rho_0) = h_{D_F}(v),$$

the spread of policy values that v produces, i.e. the support function h_{D_F} of the feature difference body $D_F := \Phi^\top (M(P) - M(P)) / (1 - \beta)$, a form we return to in §4.4. The right-hand form is two value iterations; no linear program is needed for the seminorm itself.

Proposition 1 (Properties of $\|\cdot\|_{\mathcal{D}}$). *$\|\cdot\|_{\mathcal{D}}$ is a seminorm on $\mathbb{R}^d$, and $\|v\|_{\mathcal{D}} = 0$ exactly when $v \in \mathcal{N} := \{a : a^\top \Phi^\top (\mu - \nu) = 0 \forall \mu, \nu \in M(P)\}$, the subspace of directions that add the same constant to every policy’s value. $\mathcal{N}$ contains the potential-shaping subspace and can be strictly larger. For any θ, θ' , with μ_θ^* optimal for θ and $\mu_{\theta'}$ optimal for θ' ,*

$$\text{Reg}_\theta(\pi_{\theta'}) := V_\theta(\mu_\theta^*) - V_\theta(\mu_{\theta'}) \leq \|\theta - \theta'\|_{\mathcal{D}}.$$

119 Finally $\|\cdot\|_{\mathcal{D}}$ is exactly invariant under $\phi \mapsto A\phi$, $\theta \mapsto A^{-\top}\theta$ for any invertible A .

120 The regret constant is 1, with no Cauchy–Schwarz slack and no feature-norm Lipschitz factor, because
 121 the seminorm is defined as the value spread it has to bound. It is also the only such choice, which is
 122 what licenses calling any other geometry wrong rather than merely different.

123 **Proposition 2** (Minimality). *Let N be any seminorm on $\mathbb{R}^d$ with $\text{Reg}_{\theta}(\pi_{\theta'}) \leq N(\theta - \theta')$ for all
 124 θ, θ' . Then $N(v) \geq \|v\|_{\mathcal{D}}$ for every v .*

125 *Proof.* Fix v and $\delta > 0$, and take $\theta = v$, $\theta' = -\delta v$. Any policy optimal for θ' maximizes $V_{-\delta v}$ and
 126 so minimizes V_v , while μ_{θ}^* maximizes V_v ; hence $\text{Reg}_{\theta}(\pi_{\theta'}) = \max_{\mu} V_v - \min_{\mu} V_v = \|v\|_{\mathcal{D}}$. The
 127 hypothesis with $\theta - \theta' = (1 + \delta)v$ and homogeneity of N give $\|v\|_{\mathcal{D}} \leq N((1 + \delta)v) = (1 + \delta)N(v)$.
 128 Let $\delta \downarrow 0$. $\square$

129 So $\|\cdot\|_{\mathcal{D}}$ is the pointwise smallest seminorm that bounds worst-case regret: any valid alternative is at
 130 least as large, and a strictly larger one certifies a strictly weaker statement at the same radius. The
 131 restriction to $\delta > 0$ is not cosmetic—at $\delta = 0$ every policy is optimal for $\theta' = 0$ and the construction
 132 says nothing. Since shaping directions lie in $\mathcal{N}$, replacing θ by $\theta + c$ for $c \in \mathcal{N}$ leaves both the
 133 score and the calibrated set unchanged; the angular score, computed from θ ’s normalized coordinates,
 134 has no such property. Optimal behavior cannot identify positive reward scale either, and we remove
 135 that symmetry without building quotient coordinates: for any behaviorally nonnull reward θ we set
 136 $\theta^{\circ} := \theta / \|\theta\|_{\mathcal{D}}$, and compare rewards by $\|\theta^{\circ} - \theta'^{\circ}\|_{\mathcal{D}}$, a pseudometric with range $[0, 2]$ that vanishes
 137 exactly when $\theta' = c\theta + a$ for some $c > 0$, $a \in \mathcal{N}$, and that inherits the normalized regret bound
 138 (Proposition 6).

139 4.2 Score, quantile, and what they certify

140 Write $\bar{\theta}^{\circ} = \bar{\theta} / \|\bar{\theta}\|_{\mathcal{D}}$ for the *unit-span center* and define the *span score* s_k and *span ball* B_q ,

$$s_k := \min_{\theta \in K(\pi_k)} \|\theta - \bar{\theta}^{\circ}\|_{\mathcal{D}}, \quad B_q := \{\theta : \|\theta - \bar{\theta}^{\circ}\|_{\mathcal{D}} \leq q\}.$$

141 Throughout, *coverage* names only the property $\Pr[E] \geq \gamma$ for a stated event E . The two events at
 142 issue are *intersection*, $K(\pi_{\text{new}}) \cap B \neq \emptyset$, and *containment*, $\hat{\theta}_{\text{new}}^{\circ} \in B$. Split conformal delivers
 143 coverage of whichever event the score’s sublevel set defines, and nothing more.

144 Because $\|\cdot\|_{\mathcal{D}}$ has a linear-program epigraph and $K(\pi_k)$ is polyhedral, s_k is a single LP in $d + 2|S|$
 145 variables, with no second-order cone anywhere; this is what would be lost by intersecting the cone
 146 with a Euclidean ball. Since $0 \in K(\pi_k)$ and $\|\bar{\theta}^{\circ}\|_{\mathcal{D}} = 1$, we have $s_k \in [0, 1]$, and both the score and
 147 its quantile are dimensionless and unchanged under $\bar{\theta} \mapsto c\bar{\theta} + a$ for $c > 0$, $a \in \mathcal{N}$.

148 **Proposition 3** (Intersection coverage). *Under Assumption 1, let q_{γ} be the τ -th smallest s_k among N_{val}
 149 calibration demonstrators, with $\tau = \lceil \gamma(N_{\text{val}} + 1) \rceil$ and the sentinel $+\infty$ if $\tau = N_{\text{val}} + 1$. For a fresh
 150 i.i.d. demonstrator, $s_{\text{new}} \leq q_{\gamma}$ holds if and only if $K(\pi_{\text{new}}) \cap B_{q_{\gamma}} \neq \emptyset$, and $\Pr[K(\pi_{\text{new}}) \cap B_{q_{\gamma}} \neq$
 151 $\emptyset] \geq \gamma$.*

152 The probability is the standard split-conformal rank argument [Vovk et al., 2005] applied verbatim.
 153 What belongs to this construction is the equivalence, immediate from s_k being a minimum: coverage
 154 of the score is intersection coverage of the ball, with no lossy conversion in between. The contrast
 155 is the obvious repair of the angular cap, which calibrates the value gap under $\bar{\theta}$ and converts it to a
 156 Euclidean ball by inverting a Lipschitz bound (Appendix A). That bound licenses $\text{gap} \leq \text{const} \times$
 157 distance and never the reverse, so its radius certifies neither event—a measured containment rate of
 158 0.004 in Figure 1(b).

159 4.3 From intersection to containment

160 $K(\pi)$ is not an equivalence class. Even after removing scale and null directions it can contain
 161 rewards with genuinely different policy orderings, so touching the cone says little about where the
 162 demonstrator’s reward is. Write $F^{\circ}(\pi) := \{\theta \in K(\pi) : \|\theta\|_{\mathcal{D}} = 1\}$ for the normalized cone,
 163 understood modulo the null directions.

164 Which event the quantile certifies depends only on which distance to $F^\circ(\pi)$ is scored. The latent
 165 reward lies in $F^\circ(\pi_{\text{new}})$, so scoring the *largest* distance,

$$s^{\max}(\pi) := \max_{u \in F^\circ(\pi)} \|u - \bar{\theta}^\circ\|_{\mathcal{D}},$$

166 gives $\|\hat{\theta}_{\text{new}}^\circ - \bar{\theta}^\circ\|_{\mathcal{D}} \leq s^{\max}(\pi_{\text{new}})$ pointwise, and the same split-conformal argument then certifies
 167 containment directly, under Assumption 1 alone. This is the score we want. It is also the maximization
 168 of a convex function over a nonconvex set, and we do not solve it. Our s_k scores the *smallest* distance
 169 instead: a linear program, but it bounds the wrong side. One assumption converts one into the other.

170 **Assumption 2** (Certified cone width). *There is a known $\eta \in [0, 2]$ with $\sup_{u, v \in F^\circ(\pi)} \|u - v\|_{\mathcal{D}} \leq \eta$
 171 almost surely for the deployment population, and each latent reward is behaviorally nonnull.*

172 Under Assumption 2, $s^{\max}(\pi) \leq 2s(\pi) + \eta$ pointwise: when $s(\pi) < 1$ the near witness renormalizes
 173 at a cost of $2s(\pi)$ and the cone width covers the rest, and when $s(\pi) = 1$ the bound holds trivially
 174 because $s^{\max} \leq 2$. Since $t \mapsto 2t + \eta$ is increasing it commutes with order statistics, so the pointwise
 175 inequality passes straight to the quantile, $q_\gamma^{\max} \leq 2q_\gamma + \eta =: R$, which is what Theorem 1 uses.
 176 Read this way, Assumption 2 is not information the pipeline needs in order to work: it is the price of
 177 replacing a nonconvex score by a linear program. The factor 2 renormalizes the witness, and η stands
 178 in for what an exact $s^{\max}$ would have measured. A small η can come from side information, multiple
 179 environments, or active comparisons; with none of these, the only universal certificate is $\eta = 2$.

180 **Theorem 1** (Two-stage containment and safe prescription). *Under Assumptions 1–2, let q_γ be
 181 the Stage-I quantile and set $R := \min\{2, 2q_\gamma + \eta\}$. Then $\Pr[\|\hat{\theta}_{\text{new}}^\circ - \bar{\theta}^\circ\|_{\mathcal{D}} \leq R] \geq \gamma$. Let
 182 $\mu_R \in \arg \max_{\mu \in M(P)} \min_{\theta \in B_R} [V_\theta(\mu) - V_\theta(\mu_{\text{ref}})]$ for a fixed reference $\mu_{\text{ref}} \in M(P)$; then*

$$\Pr[V_{\hat{\theta}_{\text{new}}}(\mu_R) \geq V_{\hat{\theta}_{\text{new}}}(\mu_{\text{ref}})] \geq \gamma.$$

183 The argument is short. On the Stage-I event some $\tilde{\theta} \in K(\pi_{\text{new}})$ is within q_γ of the unit-span center,
 184 so if $q_\gamma < 1$ the reverse triangle inequality makes $\tilde{\theta}$ nonnull; normalizing it moves it by at most
 185 another q_γ , which is where the factor 2 comes from, and Assumption 2 then covers the remaining
 186 distance to the latent reward inside the same cone. Minimizing over $F^\circ(\pi)$ instead of $K(\pi)$ would
 187 remove the factor 2, giving $R = \min\{2, q_\gamma + \eta\}$; maximizing rather than minimizing over $F^\circ(\pi)$
 188 removes η as well. Both are nonconvex, and both are what keeping Stage I a linear program costs.

189 4.4 The deployed policy

190 B_R is unbounded—it contains the whole affine subspace $\bar{\theta}^\circ + \mathcal{N}$ —so worst-case *absolute* value
 191 over it is unbounded below in general. The reading we take is that absolute value is the wrong
 192 criterion: a direction with no behavioral content can be scaled without bound, and only a comparison
 193 between two occupancy measures is insensitive to it. We therefore optimize worst-case value
 194 relative to a reference μ_{ref} . With $x := \Phi^\top(\mu - \mu_{\text{ref}})/(1 - \beta)$ and D_F the feature difference body
 195 $\Phi^\top(M(P) - M(P))/(1 - \beta)$,

$$\min_{\theta \in B_R} [V_\theta(\mu) - V_\theta(\mu_{\text{ref}})] = [V_{\bar{\theta}^\circ}(\mu) - V_{\bar{\theta}^\circ}(\mu_{\text{ref}})] - R\gamma_{D_F}(x),$$

196 where γ_{D_F} is the Minkowski gauge of D_F , the polar dual of $\|\cdot\|_{\mathcal{D}}$. Since $\mu - \mu_{\text{ref}}$ ranges over
 197 $M(P) - M(P)$ by construction, the penalty is automatically normalized to $[0, 1]$, R reads directly
 198 in regret units, and no sign condition can arise. The gauge has an LP epigraph, so the whole robust
 199 problem is one linear program.

200 **Proposition 4** (Degeneracy threshold). *For the unit-span center, if $R \geq 1$ the optimal relative robust
 201 value is zero, attained by μ_{ref} ; if $R > 1$ every optimizer is feature-equivalent to the reference. A
 202 nontrivial safe improvement therefore requires $2q_\gamma + \eta < 1$.*

203 Vacuity is therefore auditable: the implementation reports q_γ , η , R , and whether $R < 1$. One
 204 modelling choice remains. Centering at $\bar{\theta}$'s own optimal occupancy is degenerate, since the reference
 205 is then optimal for every R ; all experiments instead center at the calibration population's mean
 206 occupancy, which asks whether a policy beats what a random draw from the population achieves.
 207 Since Theorem 1 guarantees dominance only against whichever reference is chosen, a sensitivity
 208 analysis over that choice is required before deployment.

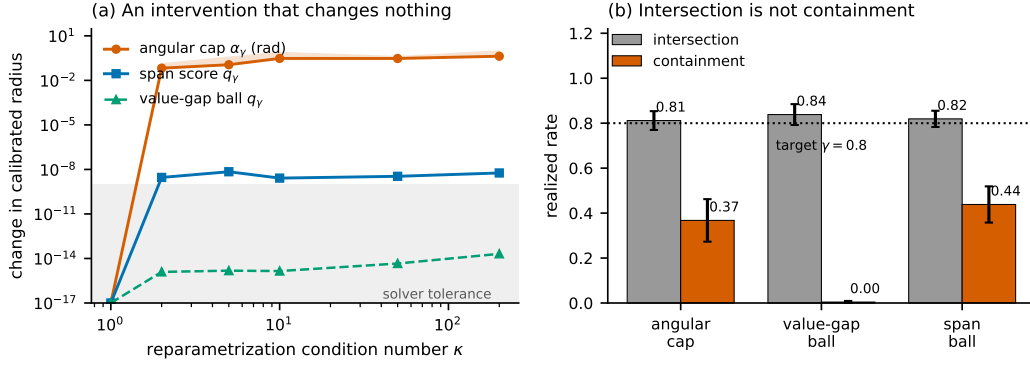


Figure 1: The two failures. **(a)** A random invertible feature reparameterization of condition number κ leaves every pointwise reward and policy unchanged, yet the inherited angular cap moves; the span score does not, up to solver tolerance. The value-gap score is also invariant—it is computed from values rather than from coordinates—and sits lower still because it needs no LP; §4.2 and Appendix A show that its converted radius nonetheless certifies neither event, so invariance alone is not sufficient. Shading spans mean to worst over 10 seeds. **(b)** At $\gamma = 0.8$ every construction attains intersection coverage at target, but containment at the same Stage-I radius falls short. The right-hand bars measure containment in the invariant span distance; the value-gap bar uses that baseline’s Euclidean conversion, which certifies neither event. Error bars are 95% t intervals over 30 environment–seed blocks.

209 5 Experiments

210 The experiments are deliberately narrow, each testing one link in the calibrate–identify–prescribe
 211 chain: whether the calibrated radius survives an intervention that changes nothing observable (E1),
 212 whether intersection coverage implies containment (E2), and how strong an external certificate Stage
 213 II would need (E3). Each isolates one mechanism rather than benchmarking end-to-end regret.

214 We use random tabular MDPs, a Voronoi-feature gridworld, and Objectworld, each with discount
 215 $\beta = 0.9$, $|S| \leq 36$, and $d \leq 5$. We fit MaxEnt IRL and Bayesian IRL by Metropolis–Hastings
 216 on 20 training demonstrators, calibrate on 20 at target level $\gamma = 0.8$, and evaluate on 20 fresh
 217 demonstrators, for 10 environment seeds: 60 runs and 1,200 test rewards. Draws within a run share
 218 a fitted center and radius, so intervals are 95% Student- t intervals over the 30 environment–seed
 219 blocks after averaging the two estimators within each block. Computation used a 20-core laptop, no
 220 GPU, and no commercial solver. An exact-fit suboptimality LP baseline is not reported: it returned a
 221 span-zero center in one of 30 fits, for which unit-span normalization is undefined, and we record that
 222 as an abstention rather than perturbing the reward by an arbitrary epsilon.

223 **E1: an intervention that changes nothing.** We draw one random invertible A at each $\kappa \in$
 224 $\{1, 2, 5, 10, 50, 200\}$ and set $\phi \mapsto A\phi$, $\theta \mapsto A^{-\top}\theta$. Coordinates change; every scalar reward
 225 $\phi(s, a)^\top \theta$ does not (maximum numerical discrepancy 2.2×10^{-14}), so a behaviorally meaningful
 226 radius should not move. At $\kappa = 200$ the angular radius changes by 0.424 radians (24.3°) on average
 227 and 1.071 radians in the worst seed, while the span radius changes by at most 2.0×10^{-8} (Figure 1a).
 228 This is a falsification test rather than a favorable data distribution: the compared problems are the
 229 same problem.

230 **E2: intersection holds, containment does not.** At $\gamma = 0.8$, pooled empirical intersection rates are
 231 0.812 ± 0.042 , 0.838 ± 0.046 , and 0.819 ± 0.036 for the angular, value-gap, and span scores; the
 232 first and third are exactly inverse-set intersection, the second only score coverage. Containment at the
 233 same Stage-I radius is only 0.368 ± 0.095 for the angular cap and 0.438 ± 0.081 for the span ball,
 234 while the value-gap-to-Euclidean conversion contains 0.004 ± 0.005 (Figure 1b). Conformal ranking
 235 behaves exactly as the theory says; containment does not follow from q_γ alone. The separation is a
 236 property of which distance is scored, not of the calibration: the same rank argument applied to $s^{\max}$
 237 would place the second bar at the target as well.

E3: how much identification is missing? The calibrated q_γ has pooled mean 0.316, median 0.289, and range $[0.019, 0.871]$, so the observable budget $b := \max\{1 - 2q_\gamma, 0\}$ is usually positive but small. A certificate must satisfy $\eta < b$ for the safe prescription to differ from the reference: with $\eta = 0$, 0.2, and 0.5, respectively 85.0%, 76.7%, and 36.7% of runs remain nontrivial, and the environment matters (at $\eta = 0.2$, 95% for random MDPs, 75% for Objectworld, 60% for gridworld). The universal certificate $\eta = 2$ gives $R = 2$ everywhere, with containment at radius R equal to 1.000, reference dominance 1.000, and nontriviality 0.000—which verifies the theorem and the implementation while deploying exactly the reference occupancy: the guarantee holds and says nothing. The full frontier is in Figure 2.

What these experiments do not show. Using the Stage-I radius directly can lower or raise true-reward regret; in the balanced six-cell audit it lowers mean regret in two cells of six and raises it in four (Table 2). That policy is not covered by Theorem 1, which uses R , and our relative objective guarantees dominance over the reference rather than over the point-estimate policy. We do not report the scale, shift, and RLHF relabeling probes here: they are exploratory and outside the claims of Theorem 1.

6 Discussion

Metric choice is real and measurable (E1), but a conformal intersection radius is not yet an uncertainty set for the latent reward (E2). Theorem 1 prices that gap and E3 makes the price observable: safe prescription needs a certified cone width below the run-specific budget $1 - 2q_\gamma$. The universal width 2 proves containment but pushes the robust solution back to the reference: safe, but with no decision content. That is a fact about what behavior can identify, not a bug in the algorithm. The open problem is therefore not another IRL point estimator, and it has two independent routes.

The algorithmic route is to compute $s^{\max}$ itself, which would need no η at all. Since $\|v\|_{\mathcal{D}}$ is a maximum of finitely many linear functionals of v —one per pair of vertices of $M(P)$ —its unit ball is polyhedral, so $F^\circ(\pi)$ is the intersection of a polyhedral cone with that boundary, a union of finitely many polyhedral pieces, and the maximum of a convex function over such a union is attained at a vertex of one of them. For small d this is enumerable; the complexity in general, and whether a useful relaxation sits between the LP and the exact program, we leave open. The data route is to certify a small η : repeated observations of one demonstrator under several dynamics, active comparisons, or externally known restrictions on the reward. The two routes are complementary—one shrinks what must be assumed, the other strengthens what can be assumed—and either moves the same frontier in Figure 2.

Our exchangeability unit is demanding—a demonstrator with a fully observed exact optimal policy in a shared known tabular MDP—and finite trajectories, approximate optimality, subgroup-conditional coverage, and MDP shift each need separate analysis. Covariate-shift weighting [Tibshirani et al., 2019] may address the statistical part of transfer but not the identifiability conditions studied by Schlaginhaufen and Kamgarpour [2024].

The closest setting we have not tested is preference-based reward modelling [Christiano et al., 2017, Rafailov et al., 2023], where the same two questions recur with a different calibration unit: one annotator, one calibration point. A Bradley–Terry reward model is identified only up to the transformations that leave every modelled comparison unchanged, so the rewards consistent with an annotator’s preferences again form a set rather than a point, and the distinction this paper is about—whether a calibrated radius merely meets that set or contains the annotator’s own reward—applies unchanged. Two things do change. The comparison set is a polyhedral cone only when the comparisons are noiseless and the model is linear in features, so $K(\pi)$ must be replaced by whatever the preference likelihood makes feasible; and annotators are not exchangeable in the way Assumption 1 needs when they are recruited in batches or disagree systematically. We raise this because a reward model trained on preferences is then optimized against by a policy, which is structurally the pipeline we have been auditing, not because our experiments say anything about it. The pattern still generalizes past IRL: whenever an inverse problem is only partially identified, a calibrate-then-prescribe pipeline must answer three questions separately—in which geometry the score is measured, how the quantile is computed, and which event the calibrated set certifies about the latent parameter—and conformal prediction answers only the second. CIO can leave the other

two implicit because in one-shot inverse optimization the inverse-feasible sets are thin; IRL is the regime where they are not.

References

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376 A Deferred constructions

377 **Proposition 5** (Resolvent-based score reduction). *Fix π ; let $P_\pi(s, s') = P(s, \pi(s), s')$, let Φ_π be*
378 *the feature matrix restricted to π ’s actions, and let $W_\pi := (I - \beta P_\pi)^{-1} \Phi_\pi$. Then $V_\theta^\pi = W_\pi \theta$ and*
379 *$Q_\theta^\pi(s, a) = \Psi_\pi(s, a)^\top \theta$ with $\Psi_\pi(s, a) := \phi(s, a) + \beta P(s, a, \cdot) W_\pi$, both linear in θ , so*

$$K(\pi) = \{\theta : \Psi_\pi(s, a)^\top \theta \leq \Psi_\pi(s, \pi(s))^\top \theta \forall s, a\}$$

380 *exactly. The angular baseline additionally uses $\hat{\Theta}_2(\pi) = K(\pi) \cap \{\theta : \|\theta\|_2 \leq 1\}$. Both have*
381 *d decision variables instead of $d + |S|$ and $|S|(|A| - 1)$ nontrivial linear inequalities, after one*
382 *resolvent solve per distinct policy.*

We verify this against a joint- (θ, V) reference to solver precision, at $2\text{--}5.5\times$ the speed at a few hundred states, and update it across demonstrators by Sherman–Morrison–Woodbury in $O(k|S|^2 + k^3)$ when policies differ at k states.

Proposition 6 (Projective span geometry). $d_o(\theta, \theta') := \|\theta^\circ - \theta'^\circ\|_{\mathcal{D}}$ is a pseudometric with range $[0, 2]$, invariant to independent positive rescaling, addition of elements of $\mathcal{N}$, and invertible feature reparameterization. For nonnull rewards $d_o(\theta, \theta') = 0$ iff $\theta' = c\theta + a$ for some $c > 0$ and $a \in \mathcal{N}$, and $\text{Reg}_\theta(\pi_{\theta'})/\|\theta\|_{\mathcal{D}} \leq d_o(\theta, \theta')$.

Theorem 2 (Concavity under reward-only ambiguity; failure with dynamics ambiguity). (a) For any nonempty compact $C \subseteq \mathbb{R}^d$, if P is fixed and known then $\max_{\mu \in M(P)} \min_{\theta \in C} \theta^\top \Phi^\top \mu$ is a concave maximization over the fixed polytope $M(P)$, hence a convex outer problem whenever C admits a tractable linear-minimization oracle. (b) The guarantee need not survive when ambiguity couples reward and transition: there is a 2-state, 2-action MDP and a convex one-dimensional non-product ambiguity set over (θ, P) whose worst-case value function is not concave.

Part (a) is why the robust step is routine once written in occupancy space, and covers the fixed-kernel regime for an arbitrary compact set. Part (b) is existential (Appendix C.2); rectangular and other structured cases remain tractable, and failure of concavity is not a hardness proof. Xu et al. [2026] instead use d -rectangular sets whose reward and transition components vary together across data sources. Part (a) does not cover that setting, since fixing P removes exactly the coupling; what it does cover is the reward-only restriction of such a set, which is one instance of the compact C .

The inherited angular baseline. For this baseline only, we replace every nonzero point estimate by its Euclidean normalization, $\|\bar{\theta}\|_2 = 1$. With the score $c_k = \max_{\theta \in \hat{\Theta}_2(\pi_k)} \theta^\top \bar{\theta}$ and $\alpha_\gamma := \arccos(q_\gamma)$, split-conformal calibration on c_k is CIO’s exchangeability argument transplanted verbatim [Vovk et al., 2005], giving $\mathbb{P}(\hat{\Theta}_{2,\text{new}} \cap C(\bar{\theta}, \alpha_\gamma) \neq \emptyset) \geq \gamma$; we omit the routine restatement. This is intersection nonemptiness, not containment. Because the zero reward is Bellman-feasible for every policy, any cap with $\alpha \geq \pi/2$ makes the intersection event automatic, so intersection coverage alone never establishes reward identification. Under the sign condition $a(x) \leq -\|x\|_2 \cos \alpha$ (writing $x = a(x)\bar{\theta} + x_\perp$), the support function of the cap reduces to $a(x) \cos \alpha - b(x) \sin \alpha$ with $b(x) = \|x_\perp\|_2$, so the robust policy solves the point estimate with a fixed-weight ℓ_2 penalty $\tan \alpha_\gamma \cdot b(\Phi^\top \mu)$, set by the conformal quantile rather than tuned.

The value-gap repair and why it is incomplete. Replacing the angular score by the suboptimality gap under θ , $c_k := V_{\bar{\theta}}(\mu_{\bar{\theta}}^*) - V_{\bar{\theta}}(\mu_{\pi_k})$, removes shaping contamination for free: shaping directions earn a constant on every $\mu \in M(P)$ (an immediate consequence of the flow-conservation constraints defining $M(P)$), so adding one to $\bar{\theta}$ leaves every c_k unchanged. But converting this score to a Euclidean ball of radius $\rho := q_\gamma/\bar{v}$ needs a Lipschitz constant $\bar{v} := \|\phi\|_{2,\infty}/(1 - \beta)$, and the bound licenses gap $\leq \text{const} \times \text{distance}$ and never the reverse, so q_γ alone certifies nothing about the set (measured containment 0.004, Figure 1b). Its robust program is at least unconditional— $\min_{\|\theta - \bar{\theta}\|_2 \leq \rho} \theta^\top x = \bar{\theta}^\top x - \rho\|x\|_2$ by Cauchy–Schwarz—but an unconditional penalty on an uncertified radius is not a certificate.

B Additional experiments

Policy regret, for completeness. Table 2 reports mean true-reward regret in the balanced six-cell audit. The Stage-I span policy beats the point estimate in two of six cells and loses in four; the certified Stage-II policy at R is worse than the point estimate everywhere, which is what Proposition 4 predicts, since a radius large enough to certify containment is large enough to pull the solution toward the reference. None of these comparisons is covered by Theorem 1, whose guarantee is dominance over μ_{ref} and not over the point estimate; we report the table so that the claim in §5 has a source, not as evidence of a performance gain.

Concentration baseline. Exact conformal calibration is tighter than a DKW bound everywhere tested; on the unbounded value-gap score DKW inflates true-reward regret $4\text{--}460\times$ over the point estimate, because the same additive slack $\sqrt{\log(2/\delta)/(2N)}$ shifts a raw-value threshold far more than a cosine in $[-1, 1]$.

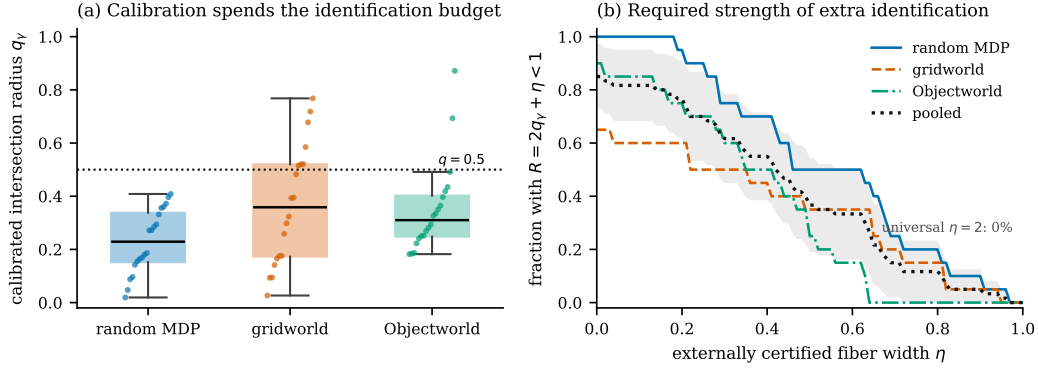


Figure 2: Identification-budget frontier. **(a)** Distribution of the dimensionless Stage-I radius over 20 runs per environment (two estimators and 10 seeds); even a zero-width cone can be nontrivial only below $q_\gamma = 0.5$. **(b)** For each hypothetical certified width η , the fraction of runs satisfying $2q_\gamma + \eta < 1$. The curve does not estimate or validate η ; it states how strong an independent identification mechanism must be. Gray shading is a 95% t interval for the pooled fraction over 30 environment–seed blocks.

Table 2: Mean true-reward regret over 10 seeds per cell, at $\gamma = 0.8$. Lower is better; bold marks cells where the span policy beats the point estimate. *Span* uses the Stage-I radius q_γ and carries no containment guarantee; *safe* uses the certified R of Theorem 1 with $\eta = 2$.

Environment	Estimator	point	span	safe (R)	angular	value-gap
gridworld	Bayesian	0.019	0.033	0.051	0.027	0.016
gridworld	MaxEnt	0.038	0.033	0.051	0.046	0.031
Objectworld	Bayesian	0.028	0.018	0.035	0.079	0.028
Objectworld	MaxEnt	0.018	0.019	0.035	0.043	0.031
random MDP	Bayesian	0.003	0.009	0.034	0.049	0.004
random MDP	MaxEnt	0.004	0.008	0.034	0.025	0.005

433 **Reproduction.** From the repository root, the main artifacts are generated by
434 experiments/run_reparam_invariance.py, experiments/run_containment_audit.py,
435 and experiments/run_center_validity.py; the figures are generated by
436 paper/make_fig1_metric.py and paper/make_fig2_identification_frontier.py.
437 All scripts set NumPy random generators from the recorded integer seed, and the audit JSON stores
438 the schema version, split sizes, target γ , environment, estimator, and seed in every row.

439 C Proofs

440 Every identity and inequality stated in the main text is also checked numerically, independently
441 of the library routines it describes, by paper/verify_formulas.py: the two-solve form of $\|\cdot\|_{\mathcal{D}}$
442 against brute-force enumeration of $M(P)$ ’s vertices, the seminorm axioms, the regret bound,
443 reparametrization invariance, the zero set $\mathcal{N}$, and the gauge form of the deployed robust objective.

444 C.1 Proof of Proposition 5

445 The value function V_θ^π solves the linear Bellman equation $V_\theta^\pi = \Phi_\pi \theta + \beta P_\pi V_\theta^\pi$. Since $\beta \in [0, 1)$
446 and P_π is stochastic, $\rho(\beta P_\pi) \leq \beta < 1$, so $I - \beta P_\pi$ is invertible via a Neumann series and
447 $V_\theta^\pi = (I - \beta P_\pi)^{-1} \Phi_\pi \theta = W_\pi \theta$. Substituting into the Bellman equation for Q_θ^π ,

$$Q_\theta^\pi(s, a) = \phi(s, a)^\top \theta + \beta P(s, a, \cdot) V_\theta^\pi = \phi(s, a)^\top \theta + \beta P(s, a, \cdot) W_\pi \theta = \Psi_\pi(s, a)^\top \theta,$$

448 which is linear in θ . By policy improvement, π is Bellman-optimal for θ iff $Q_\theta^\pi(s, a) \leq Q_\theta^\pi(s, \pi(s))$
449 for all s, a ; substituting the linear expression yields the stated cone with no relaxation, since the
450 substitution is an equality. The angular calibration score $c_k = \max_{\theta \in \hat{\Theta}_2(\pi_k)} \theta^\top \bar{\theta}$ is then a linear

451 objective over $|S||A|$ linear inequalities plus a unit-ball constraint in $\theta \in \mathbb{R}^d$ alone: an SOCP of size
 452 d , independent of $|S|$ once Ψ_{π_k} is precomputed. ■

453 C.2 Proof of Theorem 2

454 (a) Define $g_C(x) := \min_{\theta \in C} \theta^\top x$, a pointwise infimum of the affine family $\{x \mapsto \theta^\top x : \theta \in C\}$
 455 and hence concave for any nonempty C , convex or not. Composing with the linear map $\mu \mapsto \Phi^\top \mu$
 456 preserves concavity, so $\mu \mapsto \min_{\theta \in C} \theta^\top \Phi^\top \mu$ is concave in μ . Since $M(P)$ is a fixed polytope
 457 not depending on θ , the ambiguity set never touches the feasible region, only the objective, and
 458 $\max_{\mu \in M(P)} g_C(\Phi^\top \mu)$ is a concave maximization over a fixed compact convex set. This establishes
 459 the convex geometry; efficient solution also requires an efficient representation or linear-minimization
 460 oracle for C . No Bellman recursion or rectangularity assumption enters.

461 (b) Let state s_0 be active and s_1 absorbing with zero reward. At s_0 , action a_0 earns zero and moves
 462 to s_1 ; action a_1 earns $1 + t$ and returns to s_0 with probability $1 - t$ (otherwise moving to s_1), where
 463 the same ambiguity parameter $t \in [0, 1]$ couples reward and dynamics. Both actions at s_1 self-loop
 464 with zero reward. Starting from s_0 , let a stationary policy choose a_1 with probability p ; its worst-case
 465 discounted value is

$$W(p) = \min_{t \in [0, 1]} \frac{p(1 + t)}{1 - \beta p(1 - t)}.$$

466 At $\beta = 0.9$ we have $W(0) = 0$, $W(5/9) = 10/9$ (constant in t), and $W(5/18) = \min_t 10(1 +$
 467 $t)/(27 + 9t) = 10/27$, attained at $t = 0$. Since $5/18 = \frac{1}{2}(0 + 5/9)$, concavity would require
 468 $W(5/18) \geq 5/9$, but $10/27 < 5/9$. Hence W is not concave, and the outer maximization is not a
 469 convex program in this regime. ■

470 C.3 Proof of Proposition 1

471 **Seminorm properties.** Nonnegativity is immediate. Positive homogeneity: $V_v(\mu)$ is linear in v , so
 472 for $c > 0$, $\max_\mu V_{cv} - \min_\mu V_{cv} = c(\max_\mu V_v - \min_\mu V_v) = c\|v\|_{\mathcal{D}}$. Subadditivity: the supremum
 473 of a sum is at most the sum of suprema and the infimum of a sum at least the sum of infima, so
 474 $\|u + v\|_{\mathcal{D}} \leq (\max_\mu V_u + \max_\mu V_v) - (\min_\mu V_u + \min_\mu V_v) = \|u\|_{\mathcal{D}} + \|v\|_{\mathcal{D}}$.

475 **Zero set.** $\|v\|_{\mathcal{D}} = 0$ iff $V_v(\mu)$ is constant over $M(P)$ iff $v^\top \Phi^\top (\mu - \nu) = 0$ for every $\mu, \nu \in M(P)$,
 476 which is the definition of $v \in \mathcal{N}$. Every shaping direction c (via potential Φ_{pot}) satisfies $c^\top \Phi^\top \mu =$
 477 $-(1 - \beta)\rho_0^\top \Phi_{\text{pot}}$ for every $\mu \in M(P)$: expand the flow-conservation constraint $\sum_a \mu(s', a) -$
 478 $\beta \sum_{s,a} P(s, a, s') \mu(s, a) = (1 - \beta)\rho_0(s')$ and sum against Φ_{pot} . The constant cancels in $c^\top \Phi^\top (\mu -$
 479 $\nu)$, giving $S_{\text{pot}} \subseteq \mathcal{N}$. The containment can be strict: if a state s is unreachable under every policy
 480 then $\mu(s, \cdot) = 0$ throughout $M(P)$, so any v whose induced reward is supported on s lies in $\mathcal{N}$
 481 without being a potential-shaping term. We do not claim strictness in general, only that $\mathcal{N}$ is the
 482 exact zero set, which is what the invariance claims use.

483 **Regret bound.** With μ_θ^* optimal for θ and $\mu_{\theta'}$ optimal for θ' ,

$$\begin{aligned} \text{Reg}_\theta(\pi_{\theta'}) &= V_\theta(\mu_\theta^*) - V_\theta(\mu_{\theta'}) \\ &= V_{\theta-\theta'}(\mu_\theta^*) - V_{\theta-\theta'}(\mu_{\theta'}) + \underbrace{[V_{\theta'}(\mu_\theta^*) - V_{\theta'}(\mu_{\theta'})]}_{\leq 0 \text{ since } \mu_{\theta'} \text{ maximizes } V_{\theta'}} \\ &\leq \max_\mu V_{\theta-\theta'}(\mu) - \min_\mu V_{\theta-\theta'}(\mu) = \|\theta - \theta'\|_{\mathcal{D}}, \end{aligned}$$

484 the last step because $\mu_\theta^*, \mu_{\theta'} \in M(P)$.

485 **Reparametrization invariance.** For invertible A , writing $\tilde{\phi} = A\phi$ and $\tilde{v} = A^{-\top}v$, we get
 486 $\tilde{\phi}(s, a)^\top \tilde{v} = \phi(s, a)^\top v$ for every (s, a) , so the reward function is unchanged, hence $V_v(\mu)$ at
 487 every $\mu \in M(P)$, hence $\|v\|_{\mathcal{D}}$. ■

488 C.4 Proof of Proposition 6

489 On the quotient by $\mathcal{N}$, $\|\cdot\|_{\mathcal{D}}$ is a norm and every nonnull positive ray has a unit representative, so
 490 the triangle inequality, symmetry, and nonnegativity of $d_\circ$ follow from the seminorm, and $d_\circ \leq$
 491 $\|\theta^\circ\|_{\mathcal{D}} + \|\theta'^\circ\|_{\mathcal{D}} = 2$. Independent positive rescaling does not change either unit representative;

492 adding an element of $\mathcal{N}$ changes a representative only by a seminorm-zero vector; feature invariance
 493 follows from Proposition 1. If $d_o(\theta, \theta') = 0$ then $\theta/\|\theta\|_{\mathcal{D}} - \theta'/\|\theta'\|_{\mathcal{D}} \in \mathcal{N}$, and rearranging gives
 494 $\theta' = c\theta + a$ with $c = \|\theta'\|_{\mathcal{D}}/\|\theta\|_{\mathcal{D}} > 0$ and $a \in \mathcal{N}$; the converse is immediate. Finally, optimal
 495 policies are unchanged by positive normalization, so applying Proposition 1 to the two normalized
 496 rewards gives $\text{Reg}_{\theta}(\pi_{\theta'})/\|\theta\|_{\mathcal{D}} = \text{Reg}_{\theta^\circ}(\pi_{\theta'^\circ}) \leq \|\theta^\circ - \theta'^\circ\|_{\mathcal{D}}$. ■

497 C.5 Proof of Proposition 3

498 **Equivalence.** By definition $s_k = \min_{\theta \in K(\pi_k)} \|\theta - \bar{\theta}^\circ\|_{\mathcal{D}}$, so $s_k \leq q$ iff some $\theta \in K(\pi_k)$ achieves
 499 $\|\theta - \bar{\theta}^\circ\|_{\mathcal{D}} \leq q$, i.e. iff $K(\pi_k) \cap B_q \neq \emptyset$. The epigraph formulation is a feasible, bounded-below
 500 LP over a closed polyhedron, so the optimum is attained; the implementation adds a large numerical
 501 safety cap and rejects a solution if that cap is active.

502 **Intersection coverage.** $\bar{\theta}$ is a fixed measurable function of the training split, disjoint from the
 503 calibration split and the fresh demonstrator. Conditional on $\bar{\theta}$, each s_k is a deterministic measurable
 504 function of one i.i.d. demonstrator draw, so $s_1, \dots, s_{N_{\text{val}}}, s_{\text{new}}$ are exchangeable. The ascending
 505 rank of s_{new} among these $N_{\text{val}} + 1$ reals is uniform on $\{1, \dots, N_{\text{val}} + 1\}$, so $\mathbb{P}[\text{rank}(s_{\text{new}}) \leq \tau] =$
 506 $\tau/(N_{\text{val}} + 1) \geq \gamma$ by the choice of τ , and $s_{\text{new}} \leq q_\gamma$ iff $\text{rank}(s_{\text{new}}) \leq \tau$ (ties broken uniformly at
 507 random, as is standard). ■

508 C.6 Proof of Theorem 1

509 On the Stage-I event, choose a witness $\tilde{\theta} \in K(\pi_{\text{new}})$ with $\|\tilde{\theta} - \bar{\theta}^\circ\|_{\mathcal{D}} \leq q_\gamma$. If $q_\gamma < 1$, the reverse
 510 triangle inequality gives $\|\tilde{\theta}\|_{\mathcal{D}} \geq 1 - q_\gamma > 0$, and since $K(\pi)$ is a cone, $\bar{\theta}^\circ$ belongs to $F^\circ(\pi_{\text{new}})$.
 511 Moreover

$$\|\tilde{\theta}^\circ - \bar{\theta}^\circ\|_{\mathcal{D}} \leq \|\tilde{\theta}^\circ - \tilde{\theta}\|_{\mathcal{D}} + \|\tilde{\theta} - \bar{\theta}^\circ\|_{\mathcal{D}} = |1 - \|\tilde{\theta}\|_{\mathcal{D}}| + q_\gamma \leq 2q_\gamma.$$

512 The latent normalized reward lies in the same cone, so Assumption 2 and the triangle inequality
 513 give $\|\hat{\theta}_{\text{new}}^\circ - \bar{\theta}^\circ\|_{\mathcal{D}} \leq 2q_\gamma + \eta$; when $q_\gamma \geq 1$ this is subsumed by the cap $R = 2$ from the universal
 514 projective diameter. Proposition 3 assigns probability at least γ to the Stage-I event.

515 For the robust statement, μ_{ref} is feasible and has relative value zero under every reward, so the
 516 optimum at μ_R is nonnegative. On the containment event $\hat{\theta}_{\text{new}}^\circ \in B_R$, hence its relative value at μ_R
 517 is at least the worst-case relative value and therefore at least zero; positive rescaling back to $\hat{\theta}_{\text{new}}$
 518 preserves the inequality. Subtracting both values from the same optimal value gives the equivalent
 519 regret statement. ■

520 C.7 Proof of Proposition 4

521 Let $x = \Phi^\top(\mu - \mu_{\text{ref}})/(1 - \beta)$. Polar duality and $\|\bar{\theta}^\circ\|_{\mathcal{D}} = 1$ give $V_{\bar{\theta}^\circ}(\mu) - V_{\bar{\theta}^\circ}(\mu_{\text{ref}}) \leq \gamma_{D_F}(x)$,
 522 so the relative robust objective is at most $(1 - R)\gamma_{D_F}(x) \leq 0$ when $R \geq 1$. The reference has $x = 0$
 523 and attains zero. If $R > 1$, equality requires $\gamma_{D_F}(x) = 0$, i.e. the two occupancies have the same
 524 feature vector modulo the null directions; at $R = 1$ additional exposed-face ties can attain zero. ■